

ZX16 Instruction Set Test Coverage

Test Name	Instruction(s) Tested	Purpose/Behavior
R-Type Instruction Test	add, sub, slt, sltu, sll, srl, sra, or, and, xor, mv	Tests register-to-register arithmetic, comparison, shifts, and logic
I-Type Instruction Test	addi, slti, sltui, slli, srli, srai, ori, andi, xori, li	Tests immediate arithmetic, logic, shifts, and constants loading
Branch Instructions Test	beq, bne, bz, bnz, blt, bge, bltu, bgeu, j	Tests all conditional branches and jump
J-Type Instructions Test	j, jal, jr	Unconditional jump, jump and link, return
Recursive Power Function	add, sub, slt, and, or, manual stack ops	Recursion logic, stack operations, base/recursive case switching
Multiply Subroutine	add, slt, and, bne	Manual loop-based multiplication
Loop Countdown Test	add, dec, bz, j	Simple loop from 5 to 0, accumulates sum
ECALL Services Test	ecall from 1 to 10	System calls: string I/O, int I/O, audio, key, mem/register dump
Memory Test	sw, lw, sb, lb	Tests word/byte memory store/load and address loading
Arithmetic and Logic Test	add, sub, and, or, xor	Combined arithmetic and logical operations
Jump Song (Twinkle Test)	ecall audio	Plays audio tones using system calls